COURSE 2

The pentacrystals

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1. INTRODUCTION

The purpose of this talk is to bring to your attention recent new general approaches to the structure of what we call pentacrystals [1, 2, 3, 4]. A pentacrystal is any quasicrystal whose points can be written, relative to some basis $\{u_1, \ldots, u_n\}$ of a real n-dimensional Euclidean space $\mathbb{R}^n$, with coefficients in $\mathbb{Q}[\sqrt{5}]$, the quadratic extension of the rational number field $\mathbb{Q}$. In these lecture notes all quasicrystals are pentacrystals even if they do not display local 5-fold symmetry.

In [1] (see also [5]) by starting from the root lattice Q of E_8, the largest of the exceptional simple Lie algebras, and treating it as the generic "mother of all pentacrystals" — one gains an appealing unifying view of the diversity of special cases found in the literature. Indeed, the number of special cases and the avenues through which quasicrystals in dimensions 2, 3, and 4 arise from E_8 is rather large and has not yet been investigated systematically. The principal tool which allows one to achieve this is a pair of ($\mathbb{Q}[\sqrt{5}]$-)conjugate maps $\pi_{\|}$ and $\pi_{\perp}$, the conjugation being the involution of the field $\mathbb{Q}[\sqrt{5}]$ which, in particular, transforms the constants $\tau = \frac{1}{2}(1+\sqrt{5})$ and $\sigma = \frac{1}{2}(1-\sqrt{5})$ into each other. The maps are defined for our purposes in Figure 1.

In [2] a new algebraic binary operation, denoted by $\vdash$ and called quasiaddition, is defined (Section 6 below). It operates on $L_{\mathbb{F}}$, the subset of points in $\mathbb{R}^n$ with coordinates from $\mathbb{F} = \mathbb{Q}[\sqrt{5}]$. Although the true significance of this

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operation most likely exceeds the goals and aims of the study of quasicrystal, it leads here to a natural definition of a quasilattice as any set of points which has the Delaunay property (cf. Section 2) and which is closed under quasiaddition. It turns out that virtually all common quasicrystals are closed under $\vdash$ and that it is not necessary for a quasicrystal to display a 5- or 10-fold symmetry to be closed under the quasiaddition.

In addition let me also point out related work in [3, 4, 6]. More precisely in [3] the wealth of the structure of quasicrystals is made evident by (i) introducing the notion of (finite) colorings of quasicrystals, (ii) showing that there is an unlimited number of different colorings possible for each quasicrystal, and (iii) actually classifying the colorings with the property that each subset of points of a given color is closed under quasiaddition. Hence any such one-color subset is a sub-quasicrystal.

Reference [4] is a Macintosh software called *simpLie* in which the methods and operations of [1, 2, 3] are being implemented. All the examples here and in [1 − −5] were calculated and printed from *simpLie*.

In [6] the shelling problem is solved for the E_8-quasicrystals. Supposing that the origin is a quasicrystal point, one considers concentric shells containing the quasicrystal points. The problem consists in counting the number of points in each shell.

2. PRELIMINARIES

Let us recall the following facts while introducing some notation.

1. One has the real numbers $\mathbb{R}$, the rational numbers $\mathbb{Q}$ and integers $\mathbb{Z}$ as well as the extension $\mathbb{F} = \mathbb{Q}[\sqrt{5}]$ of $\mathbb{Q}$ by $\sqrt{5}$. Thus a generic element of $\mathbb{F}$ is of the form $a + b\sqrt{5}$, where $a, b \in \mathbb{Q}$. The constants $\tau = \frac{1}{2}(1 + \sqrt{5})$ and $\sigma = \frac{1}{2}(1 - \sqrt{5})$ and the identities

$$\tau + \sigma = 1, \quad \tau^2 = 1 + \tau, \quad \sigma^2 = 1 + \sigma, \quad \sigma\tau = -1, \tag{1}$$

are frequently used. Subsequently a crucial role is played by the conjugation in $\mathbb{F}$ denoted by prime:

$$x' = (a + b\sqrt{5})' = a - b\sqrt{5}, \qquad a, b \in \mathbb{Q}. \tag{2}$$

Note that $\tau' = \sigma$.

2. We will use the quaternions $\mathbb{H}_{\mathbb{F}}$ over $\mathbb{F}$. Any $x \in \mathbb{H}_{\mathbb{F}}$ can be written as

$$x = x_1 + ix_2 + jx_3 + kx_4 =: (x_1, x_2, x_3, x_4), \quad x_1, x_2, x_3, x_4 \in \mathbb{F}. \tag{3}$$

The use of quaternions is not indispensable. Superficially they serve here only as a model of a 4-dimensional Euclidean space (in the same way that the complex plane is commonly used as a model of a Euclidean plane). Indeed, one

can consider (3) as a vector in $\mathbb{R}^4$ given relative to an orthonormal basis. However, there are marvelous sets of special quaternions with properties begging to be used in the context of a study of quasicrystals; beautiful consequences and deeper insight into the problem might otherwise be obscured.

The quaternions come equipped with their standard operations:

$$\begin{aligned} \mathbb{H}\text{-conjugation of } x: &\quad \bar{x} = (x_1, -x_2, -x_3, -x_4) && (4)\\ \text{multiplication in } \mathbb{H}: &\quad xy \in \mathbb{H},\ x, y \in \mathbb{H} && (5)\\ \text{scalar product in } \mathbb{H}_{\mathbb{F}}: &\quad x \cdot y = \frac{1}{2}(x\bar{y} + y\bar{x}) \in \mathbb{F} && (6)\\ \mathbb{H}_{\mathbb{F}}\text{-norm}: &\quad N(x) = x \cdot x = |x|^2 = x\bar{x} \in \mathbb{F}. && (7) \end{aligned}$$

3. Important but little known is the rational form $(x, y)_\tau$ relative to τ on $\mathbb{H}_{\mathbb{F}}$, defined as follows. If $x \cdot y = a + \tau b$, where $a, b \in \mathbb{Q}$, then

$$(x, y)_\tau = a. \tag{8}$$

Analogously one defines a rational form relative to σ. Obviously both forms take only rational values on $\mathbb{H}_{\mathbb{F}}$.

4. Next is a special set of 120 quaternions of norm 1, called **icosians** [6]. The set I of icosians consists of the following elements of $\mathbb{H}_{\mathbb{F}}$:

$$\begin{aligned} &\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) && (9)\\ &(\pm 1, 0, 0, 0) \quad \text{\& all permutations} && (10)\\ &\frac{1}{2}(0, \pm 1, \pm\sigma, \pm\tau) \quad \text{\& all even permutations} && (11) \end{aligned}$$

The icosians form the group H_3, $|H_3| = 120$, under the quaternionic multiplication. It is isomorphic to the binary icosahedral group.

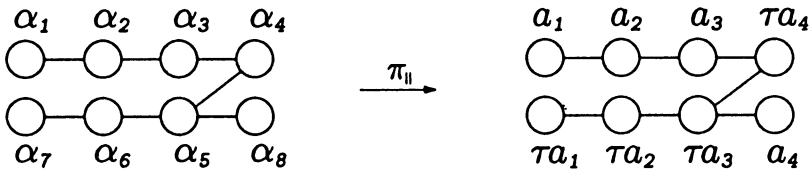
5. Finally, we use the isometric isomorphism $\pi_\parallel$ between the space $\mathbb{R}^8$ spanned by the roots of E_8 and $\mathbb{H}_{\mathbb{F}}$ [7]. See Figure 1 for its explicit definition. The mapping $\pi_\parallel$ establishes the correspondence between the root system Δ of E_8 and the icosians and their τ-multiples

$$\begin{aligned} \pi_\parallel : \Delta &\longleftrightarrow I \cup \tau I\\ (\alpha|\beta) &= 2(\pi_\parallel(\alpha), \pi_\parallel(\beta))_\tau, \qquad \alpha, \beta \in \Delta. \end{aligned} \tag{12a}$$

Similarly one defines

$$\begin{aligned} \pi_\perp : \Delta &\longleftrightarrow I \cup \sigma I\\ (\alpha|\beta) &= 2(\pi_\perp(\alpha), \pi_\perp(\beta))_\sigma. \end{aligned} \tag{12b}$$

Note that one has $(\pi_\parallel(\alpha))' = \pi_\perp(\alpha)$.



$$
\begin{aligned}
a_1 &= \tfrac{1}{2}(-\sigma, -\tau, 0, -1), & \tau a_1 &= \tfrac{1}{2}(1, -\tau^2, 0, -\tau),\\
a_2 &= \tfrac{1}{2}(0, -\sigma, -\tau, 1), & \tau a_2 &= \tfrac{1}{2}(0, 1, -\tau^2, \tau),\\
a_3 &= \tfrac{1}{2}(0, 1, -\sigma, -\tau), & \tau a_3 &= \tfrac{1}{2}(0, \tau, 1, -\tau^2),\\
a_4 &= \tfrac{1}{2}(0, -1, -\sigma, \tau), & \tau a_4 &= \tfrac{1}{2}(0, -\tau, 1, \tau^2).
\end{aligned}
$$

Fig. 1. — Mapping $\pi_{\|}$ of the simple roots of E_8 into specific icosians (quaternions). The conjugate map $\pi_{\perp}$ is obtained from $\pi_{\|}$ by replacing σ by τ and τ by σ. The Coxeter (or Dynkin) diagram of type E_8 serves here as a short way to describe a non-orthogonal basis $\{\alpha_1, \ldots, \alpha_8\}$ in an Euclidean space: Each node stands for a normalized basis vector, two nodes which are directly linked indicate the angle 120° between the corresponding basis vectors, two nodes which are not directly connected stand for a pair of mutually orthogonal basis vectors.

3. THE PENTACRYSTAL MAP

The pentacrystal map $\pi_{\|}$ is the 'projection' part of the 'cut & projection' method for construction of quasicrystals. It is useful to introduce it independently of the narrow purpose of quasicrystal construction. There is a natural generalization to dimension 4 of the $2D$ and $3D$ cases, familiar in physics and first pointed out by Elser and Sloane [13]. Once such a generalization is set up explicitly, it is extremely useful because the well known lower dimensional cases become special cases of the 4-dimensional 'mother of all pentacrystals'.

The map realizes the reduction of the orbits of the point group of the E_8-lattice (the Weyl group $W(E_8)$ of order 696 729 600 generated by eight 'elementary' reflections in the planes orthogonal to the basis vectors $\alpha_1, \ldots, \alpha_8$ defined by the diagram on Figure 1) to the orbits of the largest non-crystallographic irreducible finite group H_4 (see for example [1, 9]) acting in $\mathbb{H}_{\mathbb{F}}$ which is of order 14400.

An explicit realization of the map $\pi_{\|}$ is given in Fig. 1. There one finds the simple roots of E_8, basis of the space $V_{\mathbb{Q}}$ of all linear combinations of simple roots with rational coefficients, mapped into specific icosians a_1, a_2, a_3, a_4 and their τ-multiplies, basis of the target (quasicrystal) space $\mathbb{I}$. The rational form $(\cdot, \cdot)_\tau$ of (8) assures that the relative lengths and angles of E_8 roots are

preserved under $\pi_{\|}$. Over the real number field $\mathbb{R}$ the quasicrystal space $\mathbb{I}$ is 4-dimensional, a_i and τa_i being linearly dependent, while over the rational numbers $\mathbb{Q}$ the space is 8-dimensional because then a_i and τa_i are linearly independent.

In addition to the $\pi_{\|}$-map, we have as well its $\mathbb{F}$-conjugate, the $\pi_{\perp}$-map acting from $V_{\mathbb{Q}}$ to $\mathbb{I}$. It is obtained by replacing τ by σ everywhere in Fig. 1.

Any point X of $V_{\mathbb{Q}}$,

$$X = c_1\alpha_1 + \cdots + c_8\alpha_8, \quad c_1, \ldots, c_8 \in \mathbb{Q}, \tag{13}$$

is mapped by $\pi_{\|}$ and by $\pi_{\perp}$ as follows

$$\begin{aligned} \pi_{\|}(X) &= (c_1 + \tau c_7)a_1 + (c_2 + \tau c_6)a_2 + (c_3 + \tau c_5)a_3 + (\tau c_4 + c_8)a_4, \\ \pi_{\perp}(X) &= (c_1 + \sigma c_7)a_1' + (c_2 + \sigma c_6)a_2' + (c_3 + \sigma c_5)a_3' + (\sigma c_4 + c_8)a_4', \end{aligned} \tag{14}$$

where $a_1', \ldots, a_4'$ are the $\mathbb{F}$-conjugates of the icosians $a_1, \ldots, a_4$.

Subsequently we will need to find the length of vectors $X \in V_{\mathbb{Q}}$ and $\pi_{\|}(X)$ and $\pi_{\perp}(X)$. It is therefore useful to have the matrices of scalar products of the corresponding basis vectors:

$$((\alpha_j \mid \alpha_k)) = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 2 \end{pmatrix}, \tag{15}$$

$$((a_j \mid a_k)) = \frac{1}{2}\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -\tau \\ 0 & 0 & -\tau & 2 \end{pmatrix}, \quad ((a_j' \mid a_k')) = \frac{1}{2}\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -\sigma \\ 0 & 0 & -\sigma & 2 \end{pmatrix}. \tag{16}$$

The matrix (15) coincides with the well known Cartan matrix [10] of the Lie algebra E_8. The target space of the maps $\pi_{\|}$ and by $\pi_{\perp}$ is in $\mathbb{H}_{\mathbb{F}}$. Hence any quasicrystals constructed with the maps would be 4-dimensional. It can also be shown [1] that the underlying symmetry group is H_4. Although it is not common to speak of Cartan matrices of non-crystallographic Coxeter groups, it is justifiable [9] to call either of the matrices (16) the Cartan matrix of H_4.

Let us now single out the two important subcases of $\pi_{\|}$ and $\pi_{\perp}$ maps leading to well known icoshedral quasicrystals in 3 dimensions and to decagonal quasicrystals in 2 dimensions:

Putting $c_1 = c_7 = 0$ in (14) is equivalent to striking out the nodes of Fig. 1 labeled by α_1, α_7, and a_1, τa_1 thus reducing the dimension of the space $V_{\mathbb{Q}}$ from 8 to 6 and the E_8-diagram to that of D_6 (cf. Example 1 and Eq. (29) below). Note also that the quaternionic target space becomes the space of pure quaternions ($\bar{x} = -x$). At the same time the underlying symmetry group H_4 is reduced to its subgroup H_3, the binary icosahedral group [1].

Setting $c_1 = c_2 = c_6 = c_7 = 0$ in (14) effectively cuts from Fig. 1 the two vertical pairs of nodes on the left of the diagram. The remaining diagram is of type A_4 (cf. (28)). Resulting quasicrystals are 2-dimensional with symmetry group H_2, the dihedral group of order 10.

Among the other possible restrictions on the pentacrystal maps let us also mention the following three:

If $c_1 = c_3 = c_5 = c_7 = 0$ in (14), we have in Fig. 1 a disconnected diagram of two vertically aligned pairs opf nodes (the diagram of type $4A_1$) (cf. (28)). Resulting quasicrystals have rectangular local symmetries.

For $c_3 = c_4 = c_5 = c_8 = 0$ in (14), we have in Fig. 1 a disconnected pair of A_2 diagrams (cf. (28)), and we get quasicrystals with hexagonal symmetries.

The maps $\pi_\parallel$ and by $\pi_\perp$ for 1-dimensional quasicrystals arise by retaining only the subdiagram of (any)one pair of vertically aligned disconnected nodes in (14).

4. DEFINITION OF QUASICRYSTALS

For any fixed convex polytope $P \in \mathbb{H}$ we define a *pentacrystal* as the set Σ^P of points x,

$$x = (c_1 + \tau c_7)a_1 + (c_2 + \tau c_6)a_2 + (c_3 + \tau c_5)a_3 + (\tau c_4 + c_8)a_4, \quad c_1, \ldots, c_8 \in \mathbb{Z},$$

determined by the condition:

$$\Sigma^P = \{x \mid x' \in P\}. \tag{17a}$$

According to (14), x and x' can be *interpreted* as a result of the projections $\pi_\parallel(X)$ and $\pi_\perp(X)$ of $X \in \mathbb{R}^4$ respectively. The polytope P is called the acceptance domain of Σ^P.

Among the properties of P with important consequences for our construction are the convexity (the closure of pentacrystals under the quasiaddition of Section 7 would fail without it) and simple connectedness (to guarantee the inflation invariance (21) below). However to make things simpler, we now replace P by a ball of radius r and denote the pentacrystal by Σ^r,

$$\Sigma^r = \{x \mid N(x') \leq r^2\}, \tag{17b}$$

where $N(x')$ is the norm of x' calculated either from its quaternionic definition (7) or using the scalar product matrix in (16).

Directly from the definition follow the properties:

$$\Sigma^r \Sigma^r \subset \Sigma^{rs}; \tag{18}$$

$$\Sigma^r + \Sigma^r \subset \Sigma^{r+s}; \tag{19}$$

$$\Sigma^r \subset \Sigma^s \qquad \text{if } r < s; \tag{20}$$

$$\text{if } x \in \Sigma^r \text{ then } \tau x \in \Sigma^{r/\tau} \subset \Sigma^r. \tag{21}$$

This property is called the *inflation symmetry* of Σ^r with respect to the origin. Note that for it to be present in the general case, the acceptance domain P must be simply connected.

The quaternions can be percieved as points of a four dimensional real space. Consequently the quasicrystals of (17) are four dimensional. Our definition makes no reference to the 8-dimensional space $V_{\mathbf{Q}}$. However, in view of the maps $\pi_{\|}$ and $\pi_{\perp}$ of (14) and Fig. 1, the points x and x' of (17) can always be taken to be the result of the mappings

$$\pi_{\|}(X) = x, \quad \pi_{\perp}(X) = x', \quad X \in V_{\mathbf{Q}}. \tag{22}$$

In the 'cut and projection method', the cut is the inequality on $|x'|$ and the projection is the $\pi_{\|}$-map. Moreover, rather than choosing X from the 8-dimensional space $V_{\mathbf{Q}}$, one may consider it to belong to one of the subspaces described at the end of Section 3. In this way one gets a variety of lower dimensional quasicrystals with or without the 5-fold symmetry. Some of these options remain largely unexplored. Equally unexplored are the possibilities offered by the fact that X need not belong to a root lattice of E_8. It suffices that $X \in V_{\mathbf{Q}}$.

Finally let us underline that for a *realistic* definition of the quasicrystal an additional requirement is indispensable. The number of points which the definition (17) would have us to consider is infinite. It includes all the points of the quasicrystal and all the points falling outside of the acceptance domain. In real life one can see, and indeed there could exist, only finite quasicrystals, each occupying a volume R. Consequently the acceptance domain condition must hold only for points x in R. It is remarkable that our definition of quasicrystals is easily amendable to this requirement.

In practice one *always* works with the definition

$$\Sigma^P(R) = \{x \mid x \in R \text{ and } x' \in P\}, \tag{23}$$

where P and R are given convex simply connected domains.

The symmetry between P and R in (23) is evident: there are two quasicrystals simultaneously defined: points x in R, (quasicrystal $\Sigma^P(R)$) and points x' in P (quasicrystal $\Sigma^R(P)$). It is often convenient to choose P and R of the same shape if not of the same size. Both quasicrystals coincide when $P = R$.

It is conceivable that the duality between $\Sigma^P(R)$ and $\Sigma^R(P)$, and through it the finite size of real life quasiperiodic materials, is of fundamental nature.

5. PHASONS

The origin, $x = 0$, is a point of the quasicrystals (17) and (23). As a consequence [1] one gets the global H_4-symmetry of the quasicrystals. That symmetry disappears when the origin is shifted off quasicrystal points.

Let $\Phi \in V$ be a fixed vector which we call *phason*, and let us generalize the quasicrystal definition (17a) using the projections $\pi_{\|}(\Phi)$ and $\pi_{\perp}(\Phi)$ as follows.

$$\Sigma^r_\Phi = \left\{x + \pi_{\|}(\Phi) \mid \text{ and } N(x' + \pi_{\perp}(\Phi)) < r^2\right\}. \tag{24}$$

Clearly Σ^r_0 of (24) is Σ^r of (17b). The acceptance domain of Σ^r_Φ for $\Phi \neq 0$ is not centered at the origin anymore. The fixed point of the H_4-symmetries is now $\pi_{\|}(\Phi)$ and, in general, it does not belong to Σ^r_Φ.

6. QUASIADDITION

The quasiaddition is a recently defined binary operation [2] in the quasicrystal space whose importance to the theory of quasicrystals cannot be exaggerated. In particular all our pentacrystals turn out to be closed under it.

Suppose $L_{\mathbb{F}}$ is a linear space over $\mathbb{F}$ of dimension $n < \infty$. The quasiaddition in $L_{\mathbb{F}}$ is defined by

$$x \vdash y = \tau^2 x - \tau y, \quad x, y \in L_{\mathbb{F}}. \tag{25}$$

The non-commutativity and non-associativity of quasiaddition can be verified using the definition in a straightforward way for all $x, y, u \in L_{\mathbb{F}}$ as well as the following properties:

$$T_x(x) := x \vdash x = x; \tag{26a}$$
$$x \vdash (x \vdash y) = x \vdash y; \tag{26b}$$
$$T_x T_y^2 = T_y T_x^2; \tag{26c}$$
$$(x + u) \vdash (y + u) = (x \vdash y) + u. \tag{26d}$$

The closure of Σ^P under $\vdash$ is equally easy to check. The acceptance domain condition,

$$P \ni (x \vdash y)' = \tau'^2 x' - \tau' y' = (1 + \tau')x' - \tau' y', \tag{27}$$

turns into the convexity condition for all $x, y \in \Sigma^P$. Indeed, the coefficients of x' and y' are both positive and their sum is 1. Consequently for $(x \vdash y)'$ to be in P, the convexity of P is sufficient.

The emergence of this operation immediately suggests many interesting questions. In order to finish let me list some of them:

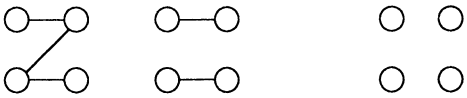
- What is the structure of a set of seed points from which repeated application of quasiaddition produces a 'reasonable looking' quasicrystal in a space of given dimension $n \geq 1$?
- What can one say about the Fourier image of the orbits of quasiaddition from the previous point?
- Description of equivalent sets of seed points. What are the minimal sets of seed points for each n?
- Study of quasicrystals in terms of the properties of the seed points.

- How does one decompose a given quasicrystal into orbits of quasiaddition (corresponding to some minimal sets of seed points)?
- Can one invent local growth mechanisms using quasiaddition?
- Using a union of several quasiaddition orbits corresponding to different 'atoms', what new structures can arise?
- What identities other than (26c) are there for the operators T_x?

7. EXAMPLES

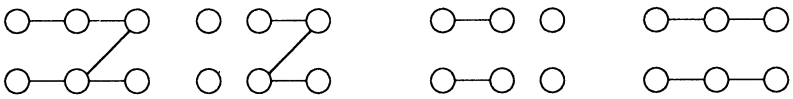
In order to be able to draw examples in a plane, we are limited to quasicrystals in 2 dimensions.

EXAMPLE 1. We thus consider the subdiagrams of the E_8-diagram of Fig. 1 consisting of two pairs of vertically aligned nodes. There are three different diagrams which one obtains. Using the standard nomenclature in Lie theory, these are the diagrams of types



$$A_4, \qquad A_2 + A_2, \qquad A_1 + A_1 + A_1 + A_1. \tag{28}$$

For quasicrystals in $3D$, one would have the following subdiagrams to consider:



$$D_6, \qquad A_4 + A_1 + A_1, \qquad A_2 + A_2 + A_1 + A_1, \qquad A_3 + A_3. \tag{29}$$

EXAMPLE 2. To each of these diagrams corresponds a root system of the Lie algebra of that type. Its root system is a finite subset of points in the root lattice Q of E_8, and its root lattice is a (root) sublattice of Q. Using the sublattices of (28) or (29) in our definition of quasicrystal, we get quasicrystals of dimension 2 or 3 of the type indicated by the corresponding diagram.

In this example let us see the effect of the mapping $\pi_{\|}$ on the three root systems of (28). The results in Fig. 2 indicate the type of local symmetry displayed by the quasicrystals of the same type. The symmetry in the case of A_4, $2A_2$, and $4A_1$ is respectively 10-, 6-, and 4-fold.

EXAMPLE 3. Examples of $2D$ quasicrystals (17) of the three types (28) are found in Fig. 3. Note the global rotational symmetry around the origin indicated by the cross.

EXAMPLE 4. The A_4-quasicrystal of the previous example is shown in Fig. 4. The origin is indicated by a cross. This time there is a non-zero shift Φ of the origin. In one case the origin is shifted to a point of the weight lattice of A_4 which does not belong its root lattice. The basis of the weight lattice are the

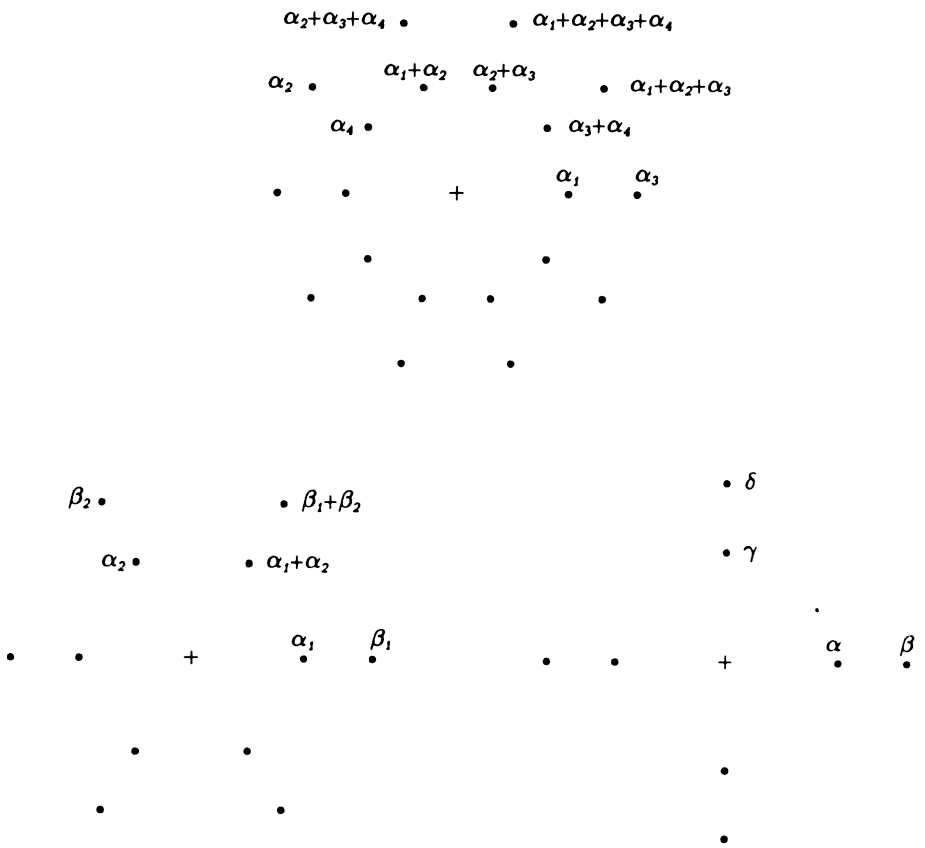


Fig. 2. — The root systems of A_4, $2A_2$, and $4A_1$ after the projection $\pi_{\|}$. The E_8 simple roots of Fig. 1 are renamed according to the conventions usual for each subalgebra. Only positive roots are labeled.

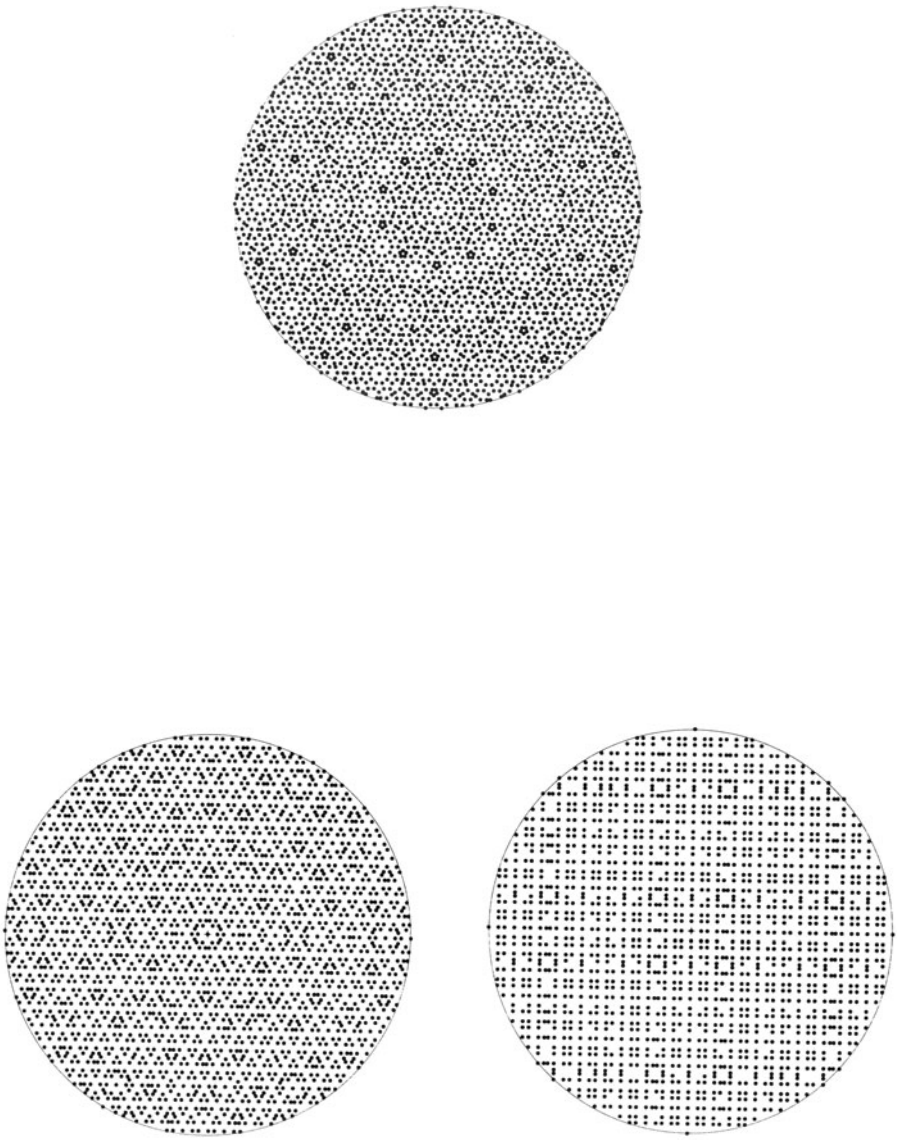


Fig. 3. — Circular window view of quasicrystals Σ^{10} of types A_4, $2A_2$, and A_1 respectively.

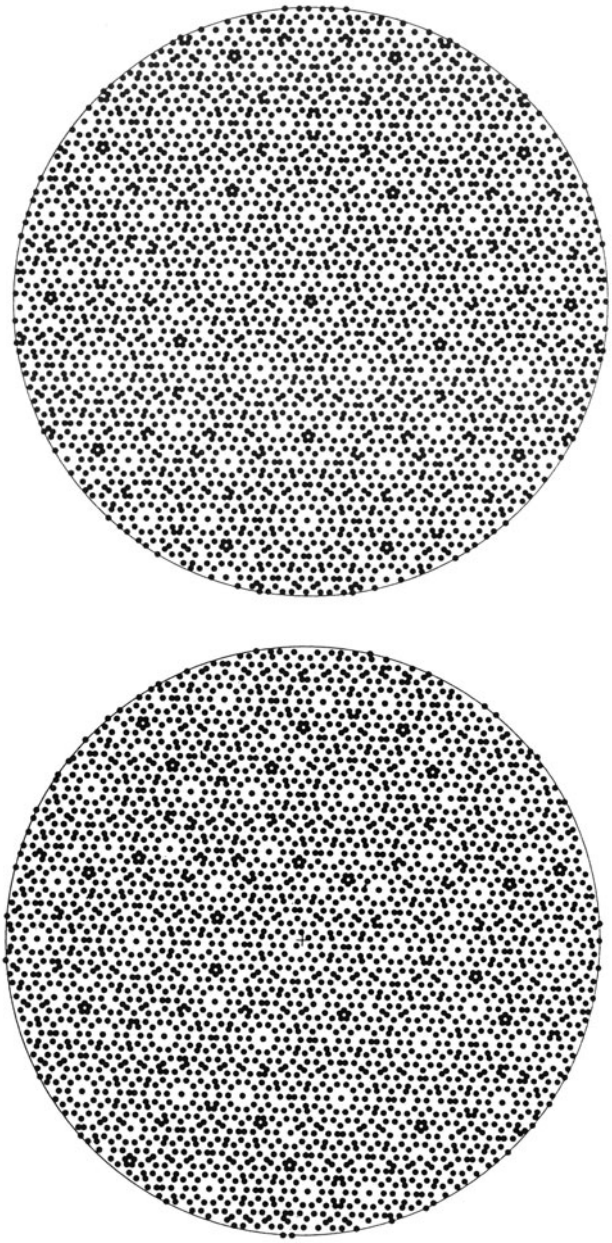


Fig. 4. — Circular window view of A_4-quasicrystals Σ^{10} with $\Phi = \omega_1 + \omega_2 + \omega_3$ and $\Phi = 0.7\omega_1 + 0.3\omega_2 + 0.2\omega_3 + 0.4\omega_4$. Note the 5-fold rotational symmetry around the origin in the first case and its absence in the second one.

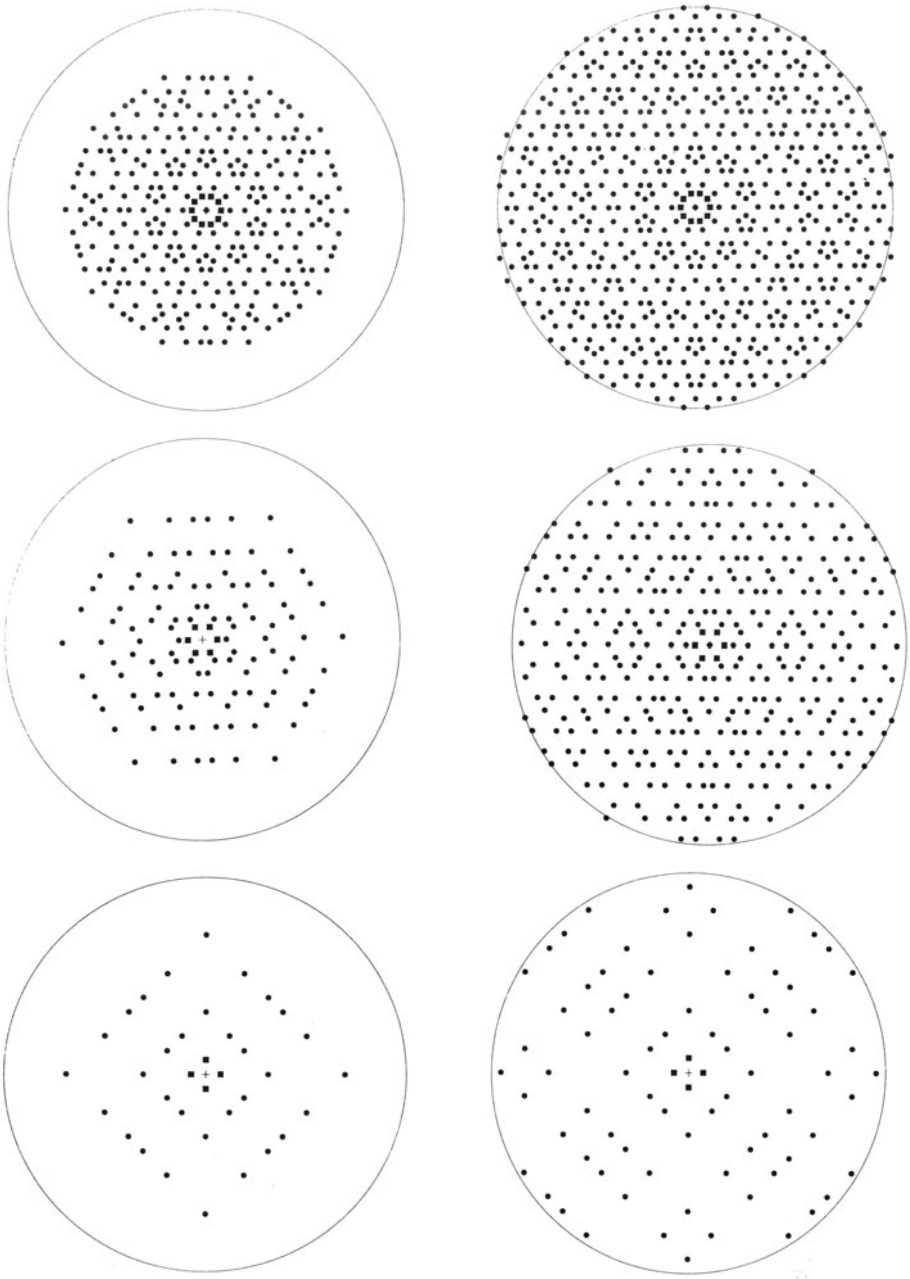


Fig. 5. — Three examples of generation of a quasicrystal by quasiaddition (25) are shown. The seed points (black squares) are vertices of regular polygons: decagon, hexagon and square. Two stages of the generation are shown for each of the examples.

fundamental weights $\omega_1, \ldots, \omega_4$. Note the reduction of the global rotational symmetry around the origin to a 5-fold one. In the second case the origin is in a generic point which not in the weight lattice. There is no global rotation symmetry left.

EXAMPLE 5. In Fig. 5 we show two stages of multiple quasiaddition for three sets of seed points: vertices of a regular decagon, hexagon, rectangle. The seed points are indicated by black squares. The generation proceeds by quasiaddition of vectors connecting the origin and the seed points to the known points of the quasicrystal according to (25).

The examples of this notes were calculated and printed using version 2 of the software [4].

As a final remark let me point out a new mechanism for generation of quasicrystals [13] in essentially any dimension, which is unlike any previously proposed method. It is based on the results of Y. Meyer [14, 15] and the *amenability* property of certain polytopes [13, 16]. The process begins with the initial seed cluster of sites from which growth proceeds radially outwards. In its simplest formulation, which can already produce an infinite variety of quasicrystals, the only global information is supplied by the choice of one or several constants determining a continuous background field for the growth process, which then proceeds locally by attempting to mimic this field. In scope the process can generate most of the possibilities offered by the 'cut and project' method, but has the advantage of avoiding its awkward non-physical aspects. More precisely we have a notion of an ideal local configuration (*ilc*) of points, for example vertices of a icosadodecahedron, or the decagon in $2D$. Starting from a finite seed (for example, a single point), each existing point of the growing quasicrystal views itself as a center of a potential *ilc*. Exactly which points of the *ilc* can actually be accepted as new quasicrystal points depends on how well some local phase conditions match some globally imposed continuous background field.

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